

Affine Transformations

An affine transformation is a transformation $T: R^2 \rightarrow R^2$, defined by

$$T(\vec{x}) = A\vec{x} + \vec{a} ; \quad \forall x \in R^2$$

For some $a \in R^2$, where A is 2×2 invertible matrix i.e. $AA^{-1} = I$.

Remarks:

1. Every orthogonal matrix is invertible but an invertible matrix may or may not be orthogonal.
2. Euclidean geometry is a subset of affine geometry or Affine transformations are the generalization of Euclidean transformation.

Types of Affine transformation:

1- Scaling, 2- Stretching, 3- Shearing

1- Scaling

A transformation $T: R^2 \rightarrow R^2$, defined by:

$$T(\vec{x}) = A\vec{x}; \quad \forall x \in R^2$$

Where

$$A = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$$

Is said to be a scaling by the factor k .

Note: 1. If $k > 1$, then its dilation/enlargement.

2. If $0 \leq k \leq 1$, then its contraction.

Example 1: (Scaling of Square)

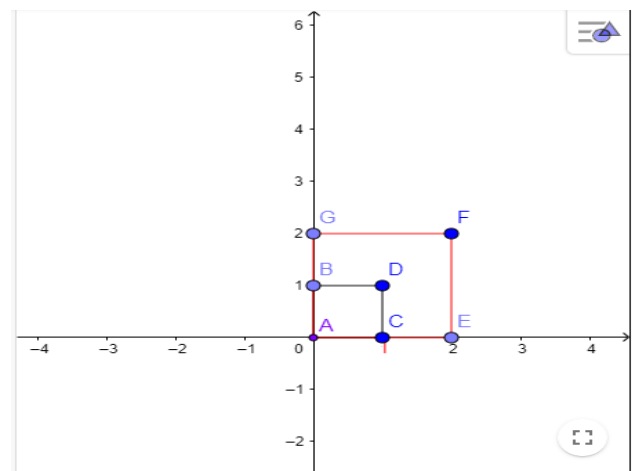
A square with vertices $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$. Scale this square by factor 2.

Solution: As

$$T(\vec{x}) = A\vec{x}: \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$



$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Example 2: (Scaling of Line)

Let $y = 2x + 3$ be a line. Determine the image of this line under the scaling by factor 3.

Solution:As

$$T(\vec{x}) = A\vec{x}: \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

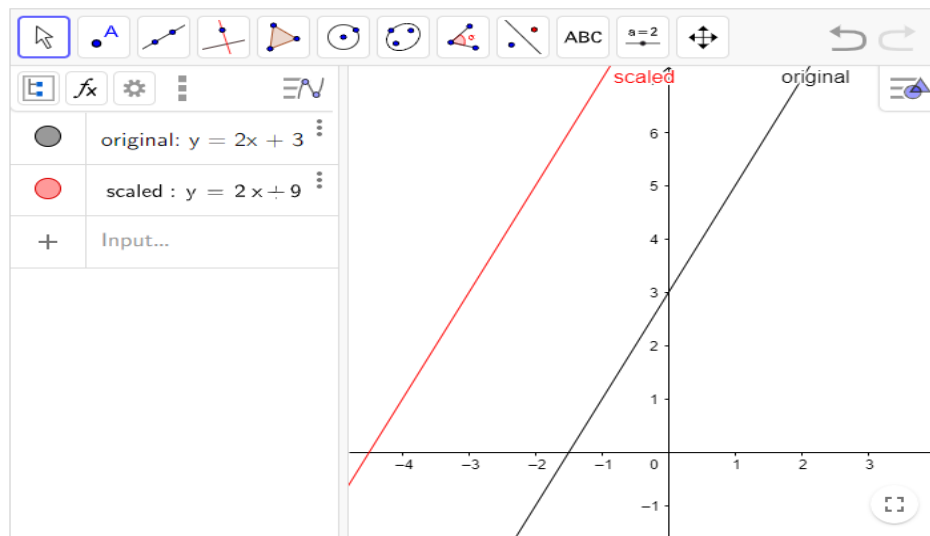
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\begin{cases} x' = 3x \\ y' = 3y \end{cases} \Rightarrow \begin{cases} x = \frac{x'}{3} \\ y = \frac{y'}{3} \end{cases}$$

Put value of x and y in original equation of line: $y = 2x + 3$,

And get the equation of scaled line with factor 3.

$$y' = 2x' + 9$$



Example 3: (Scaling of Circle)

Determine the image of the circle

$$(x - 2)^2 + (y - 3)^2 = 4$$

under the scaling by factor 2.

Solution: As

$$T(x) = Ax: \forall x \in \mathbb{R}^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

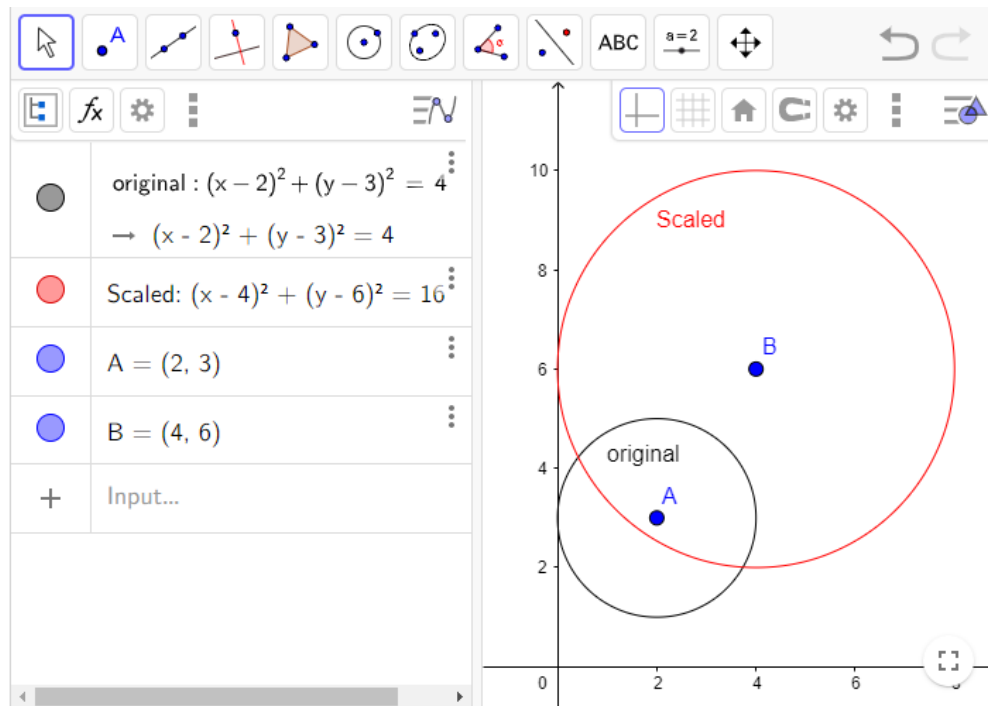
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\begin{cases} x' = 2x \\ y' = 2y \end{cases} \Rightarrow \begin{cases} x = \frac{x'}{2} \\ y = \frac{y'}{2} \end{cases}$$

Put value of x and y in original equation of circle: $(x - 2)^2 + (y - 3)^2 = 4$, implies

$$(x' - 4)^2 + (y' - 6)^2 = 16$$

This is the equation of circle by factor 2.



Example 5: (Scaling of an Ellipse)

Determine the image of an ellipse

$$\frac{x^2}{4^2} + \frac{y^2}{2^2} = 1$$

under the transformation of scaling by factor 2.

Solution: As

$$T(x) = Ax: \forall x \in \mathbb{R}^2$$

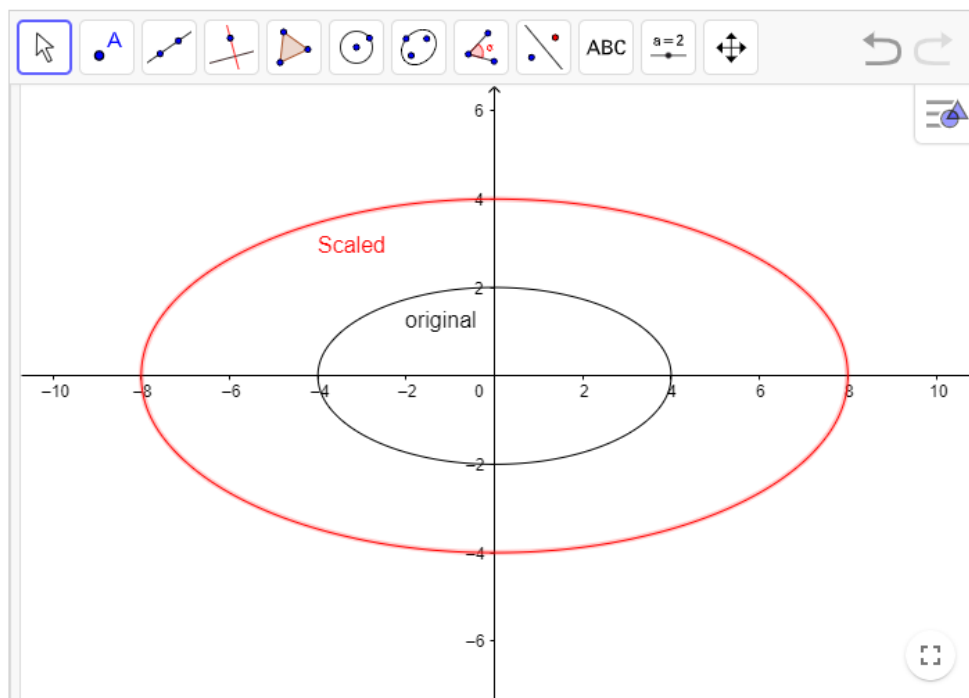
$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\begin{cases} x' = 2x \\ y' = 2y \end{cases} \Rightarrow \begin{cases} x = \frac{x'}{2} \\ y = \frac{y'}{2} \end{cases}$$

Put value of x and y in original equation of ellipse and get an ellipse scaled by factor 2.

$$\frac{x'^2}{64} + \frac{y'^2}{16} = 1$$



Practice Problems

Q1. Find the Image of a triangle with vertices $A = (2, 2)$, $B = (4, 3)$, $C = (5, 5)$, when these are scaled by factor 2.

Q2. Find the Image of line $y = 2x + 5$ under the transformation of scaling by factor 3.

Q3. Find the image of an ellipse $\frac{(x-2)^2}{4} + \frac{(y-3)^2}{9} = 1$ under the transformation of scaling by factor 2.

2-Stretching

A transformation $T: R^2 \rightarrow R^2$ defined by,

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

Is said to be stretching along x-axis by factor k if

$$A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

And along y-axis by a factor k if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

Example 1: (Stretching of a square)

Find the image of a square with vertices $A(0,0), B(1,0), C(1,1), D(0,1)$, when it is stretched along

- X-axis with factor 3.
- Y-axis with factor 2.

Solution: (along x-axis) As $T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Along y-axis

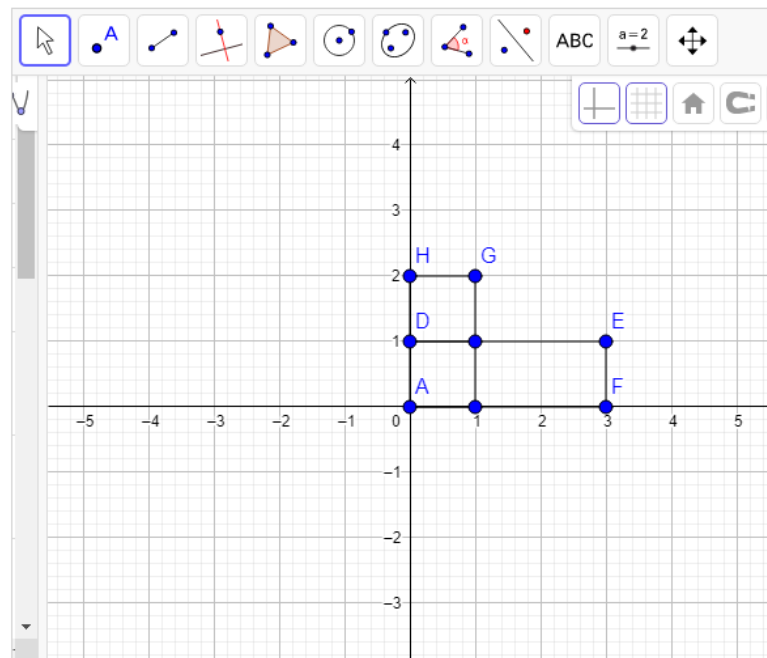
As $T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 0 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Example 2: (Stretching of triangle)

Find the image of a triangle with vertices $A(1, 1), B(4, 1), C(3, 4)$, when it is stretched along

a) X-axis with factor $\frac{3}{2}$.

b) Y-axis with factor $\frac{1}{2}$.

Solution: (along x-axis) As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$A = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix}$$

$$T \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$T \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{9}{2} \\ 4 \end{bmatrix}$$

Along y-axis: As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix}$$

$$T \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2} \end{bmatrix}$$

$$T \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Example 3: (Stretching of Circle)

Let $(x - 2)^2 + (y - 1)^2 = 4$ be a circle. Find its equation and image under stretching along x-axis by factor 2.

Solution: As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2x \\ y \end{bmatrix}$$

$$\begin{cases} x' = 2x \\ y' = y \end{cases} \Rightarrow \begin{cases} x = \frac{x'}{2} \\ y = y' \end{cases}$$

Put value of x and y in original equation of circle and get

$$\frac{(x' - 4)^2}{16} + \frac{(y' - 1)^2}{4} = 1$$

It represents the ellipse.

Practice Problems

Q1. Let $(x - 2)^2 + (y - 1)^2 = 4$ be a circle. Find its equation and image under stretching along Y-axis by factor $\frac{3}{2}$.

Q2. Find the stretching of an ellipse $\frac{(x-4)^2}{16} + \frac{(y-1)^2}{4} = 1$ along x-axis by factor 2.

Q3. Find the image of a triangle with vertices $A(1, 1)$, $B(4, 2)$, $C(2, 3)$, when it is stretched along

- a) X-axis with factor 3
- b) Y-axis with factor 1.

3- Shearing

Shearing of a point along x-axis by factor k can be defined as:

The movement of point parallel to x-axis keeping the distance of the point from the x-axis.
i.e. the movement of point parallel to x-axis depending upon the y-component of the point.

- Shearing at a point along x-axis by factor k in the matrix form can be represented as

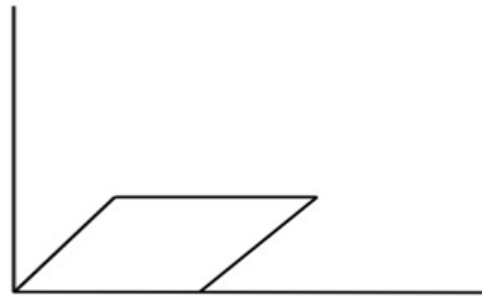
$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

Where

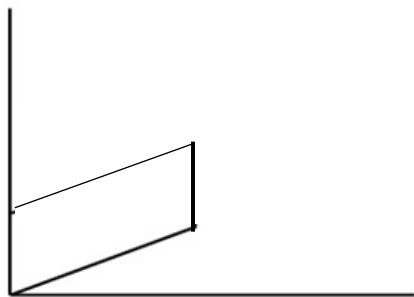
$$A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$



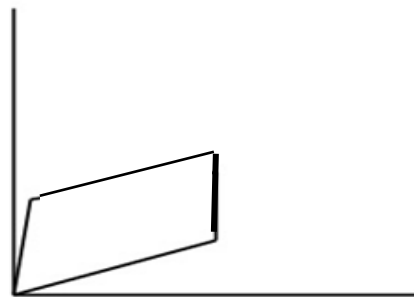
Original Object



Shear in X direction



Shear in Y direction



Shear in both directions

Example 1: (Shearing of vector)

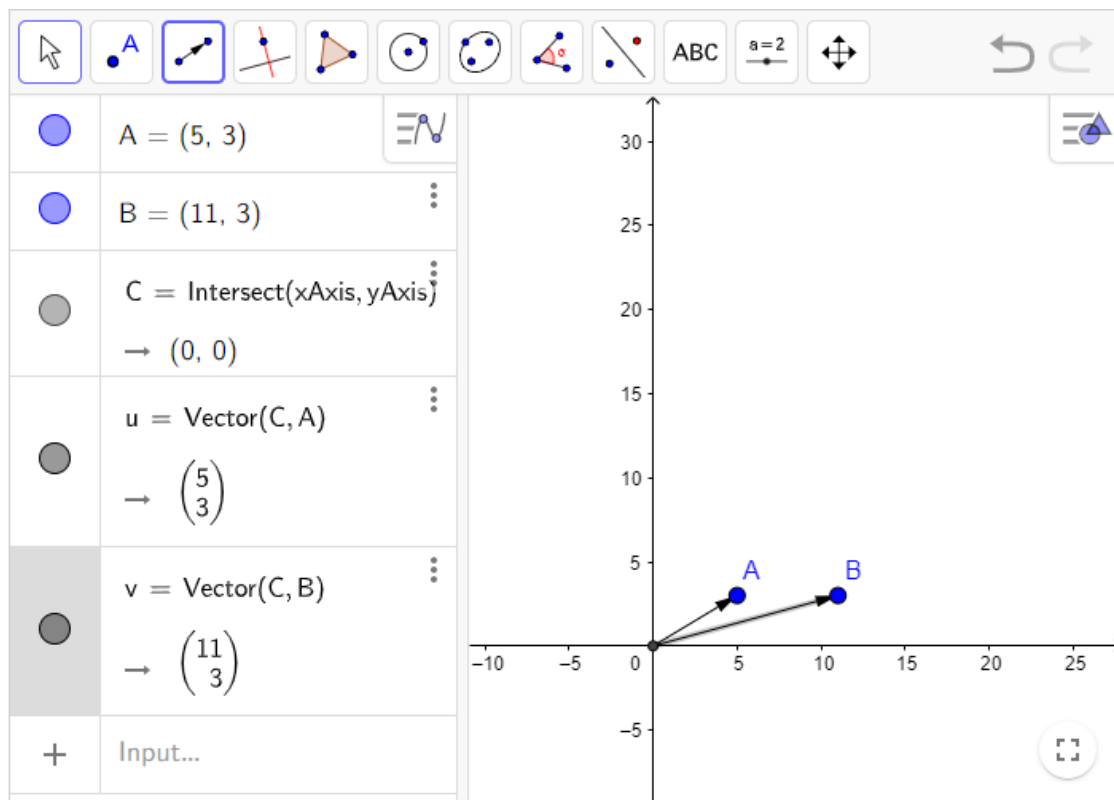
Shear a point $A = (5,3)$ along x-axis by the factor 2.

Solution: As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ky \\ y \end{bmatrix}$$

$$T \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 3 \end{bmatrix}$$



Note:

1. A transformation that slants the shape of an object is called the shear transformation.
2. Shearing of an object along y-axis by factor k in matrix form can be expressed as:

$$T(\vec{x}) = A\vec{x}; \quad \forall x \in R^2$$

Where

$$A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

Example 2: (Shearing of Triangle)

A triangle with vertices (0, 0), (4, 2), (1, 6). Find the image of triangle when it will be shear along y axis by factor 2.

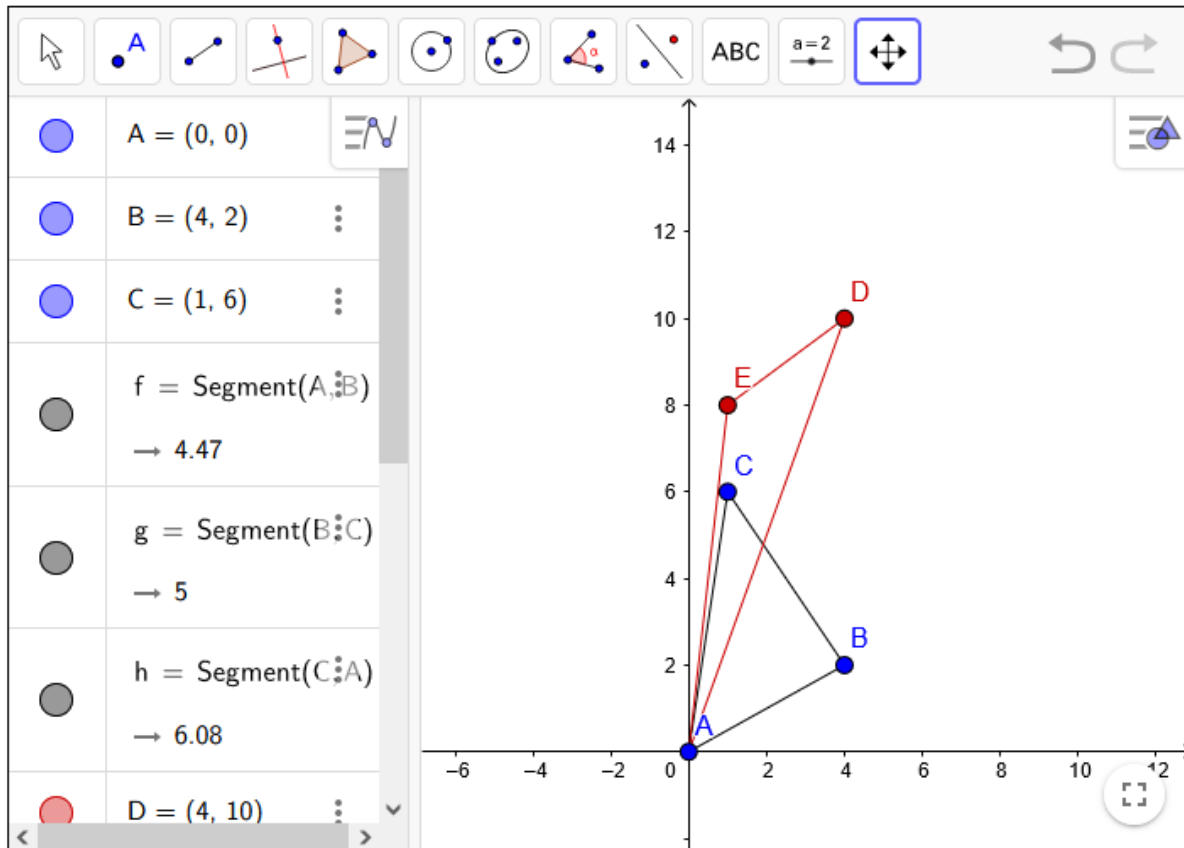
Solution: As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$



Example 3: (Shearing of Rectangle)

Consider the rectangle with vertices $(0, 0)$, $(0, 2)$, $(4, 0)$, $(4, 2)$. Shear this rectangle along x-axis by factor 2.

Example 4: (Shearing of Rectangle)

Consider the rectangle with vertices $(0, 0)$, $(0, 2)$, $(4, 0)$, $(4, 2)$. Shear this rectangle along y-axis by factor -3.

Example 5:

In each part describe the matrix operator corresponding to A and show its effect on unit square

a) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, b) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, c) $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

Example 6: Let $x^2 + y^2 = 4$ be a circle. Find its equation and image under the effect of shear parallel to x-axis by factor -2.

Solution: As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in \mathbb{R}^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x - 2y \\ y \end{bmatrix}$$

$$\begin{cases} x' = x - 2y \\ y' = y \end{cases} \Rightarrow \begin{cases} x = x' + 2y' \\ y = y' \end{cases}$$

Put value of x and y in original equation of circle: $x^2 + y^2 = 4$

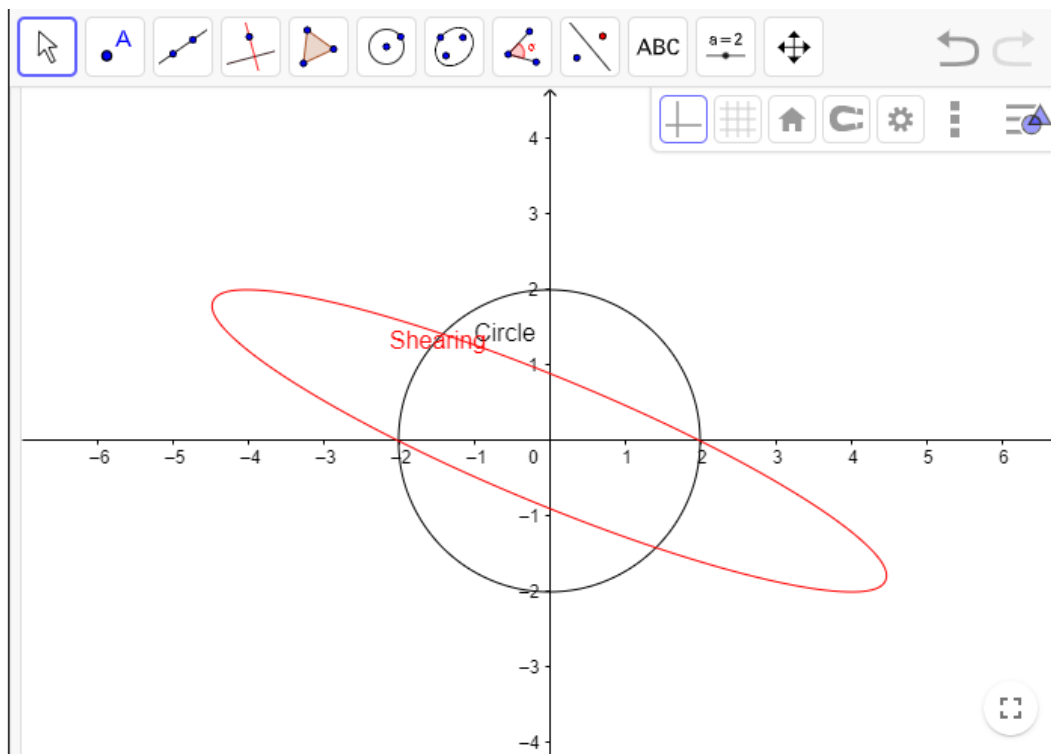
We get sheared equation of circle as (an ellipse)

$$x'^2 + 5y'^2 + 4x'y' - 4 = 0$$

Remarks: General equation of Conics:

$$Ax^2 + Bxy + cy^2 + Dx + Ey + F = 0$$

1. If $B^2 - 4AC < 0$, we get ellipse.
2. If $B^2 - 4AC > 0$, we get hyperbola.
3. If $B^2 - 4AC = 0$, we get parabola.



Example 6: Let $(x + 2)^2 + (y + 1)^2 = 4$ be a circle. Find its equation and image under the effect of shear parallel to y -axis by factor $\frac{3}{2}$.

Solution:As

$$T(\vec{x}) = A\vec{x}; \quad \forall \vec{x} \in R^2$$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ 3x + y \end{bmatrix}$$

$$\begin{cases} x' = x \\ y' = \frac{3}{2}x + y \end{cases} \Rightarrow \begin{cases} x = x' \\ y = y' - \frac{3}{2}x' \end{cases}$$

Put value of x and y in original equation of circle:

$$(x + 2)^2 + (y + 1)^2 = 4$$

$$\frac{13}{4}x^2 + y^2 - 3xy - 3x + 2y + 1 = 0$$

On Comparing with General equation of Conics, we get:

$$A = \frac{13}{4}, B = -3, C = 1$$

$$B^2 - 4AC = (-3)^2 - 4\left(\frac{13}{4}\right)(1) = 9 - 13 = -4 < 0$$

So we get ellipse.

Example 7: Let $(x - 2)^2 + (y - 1)^2 = 4$ be a circle. Find its equation and image under the effect of shear parallel to x-axis by factor 2.